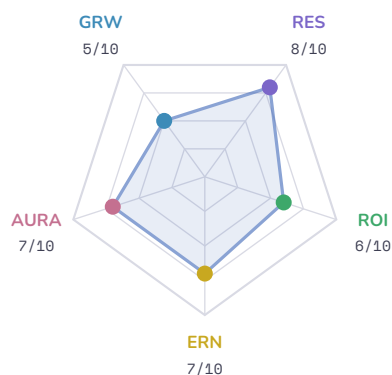


Rowan

Residency: In-state, IN

Purdue University · Mechanical Engineering · As of May 6, 2025

Mechanical Engineering at Purdue: low risk on four of five career factors, including AI displacement. Projected ROI positive by year 7.



ERN	7/10	Earning Power
ROI	6/10	Return on Investment
RES	8/10	AI Resilience
GRW	5/10	Growth Outlook
AURA	7/10	Brand Gravity

COST & ROI

4-year cost	Modeled debt	Year-1 median earnings	Break-even
\$43,200	\$28,000	\$72,000	Year 7

CAREER RISK PROFILE

Risk Factor	Level	Context
AI Displacement	LOW	Karpathy composite: 22nd percentile for automation exposure.
Debt Burden	ELEVATED	Modeled debt-to-income ratio: 0.39, above the 0.30 threshold.
Job Market Outlook	MODERATE	BLS projects 2% growth through 2032; below engineering average.
Burnout Risk	LOW	O*NET: low time-pressure and conflict demands in this role.
Earnings Ceiling	MODERATE	75th-percentile wage: \$99K; strong but not exceptional for ME.

SUGGESTED SKILLS

Bring this list to your admissions counselor — these are the skills the data says will lift your outcomes.

AI-RESILIENCE SKILLS

• Python for engineers

Lifts AI displacement resilience — exposes automation-resistant work patterns.

Coursework	Clubs / orgs	Internship / cert
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Ask: "Does Purdue Mechanical Engineering require or offer Python beyond MATLAB?"

• Machine learning fundamentals

Strengthens the technical foundation against AI-assisted design workflows.

Coursework	Clubs / orgs	Internship / cert
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Ask: "Are ML electives available in the ME curriculum at Purdue?"

CAREER-LAUNCH SKILLS

• Internship at a manufacturer

Establishes industry network and first-year earnings trajectory.

Coursework	Clubs / orgs	Internship / cert
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Ask: "What co-op and internship placement rates does Purdue ME report?"

• Technical writing certification

Communication skill differentiator for senior-track roles.

Coursework	Clubs / orgs	Internship / cert
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Ask: "Does the ME program include a technical communication requirement?"

EARNINGS-CEILING SKILLS

• Project management fundamentals

Unlocks team-lead roles that are the fastest path to \$100K+.

Coursework	Clubs / orgs	Internship / cert
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Ask: "Are project management credentials available through Purdue's engineering school?"

• Professional Engineer (PE) exam prep

PE licensure raises the earnings ceiling significantly in civil/mech fields.

Coursework	Clubs / orgs	Internship / cert
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Ask: "What is Purdue's first-attempt PE exam pass rate for ME graduates?"

QUESTIONS & FOLLOW-UPS

ASK THE COLLEGE

- Which majors at Purdue will yield careers as a Mechanical Engineer?
- How can I augment my education here with the suggested skills above?
- What percentage of Purdue ME graduates are employed in engineering roles within 6 months?

ASK YOUR PARENTS

- Is the \$28,000 in projected debt manageable on a \$72,000 starting salary?
- How does Purdue's engineering reputation compare to state-school alternatives nearby?
- What internship support does the school provide, and when should we expect one?

ASK YOURSELF

- Will I stay motivated through four years of calculus, physics, and lab work?
- Am I drawn to the physical, hands-on problem-solving that mechanical work requires?
- Do I want to stay in Indiana after graduation, or am I open to relocating?

GLOSSARY

CIP	Federal program code (Classification of Instructional Programs).	RES	AI Resilience: automation resistance score (Karpathy + Anthropic).
SOC	Federal occupation code (Standard Occupational Classification).	GRW	Growth Outlook: projected job growth through 2032 (BLS).
ERN	Earning Power: first-year earnings potential, scaled 1–10.	AURA	Brand Gravity: institution-level reputational ceiling.
ROI	Return on Investment: payoff vs. 4-year cost, scaled 1–10.	Career risk	Stress test against a career risk: AI, debt, market, burnout, ceiling.

